Module 8 — Analytical Summary

Grad Cafe Applicant Data: Statistical Findings and Interpretation

1. Average GRE score of applicants accepted into a PhD program

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Mean GRE = 327.30 (n = 322)
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This value reflects the combined GRE score (Verbal + Quantitative) among applicants with outcome = Accepted and Degree = PhD, after excluding out-of-range sentinel values (e.g. a placeholder value of 999 used in the source data to mean "not reported") that would otherwise have been miscounted as real scores. The sample size (n = 322) is smaller than an uncorrected calculation would suggest, since GRE is missing for roughly 92% of all records and a further set of entries had to be excluded as implausible placeholders rather than genuine scores — but the resulting average is now within the real GRE combined-score range (260–340) and is far more trustworthy as a result.

2. Average GPA of applicants accepted into a Master's program

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Mean GPA = 3.7166 (n = 3,929)
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This is a substantially larger and more reliable sample than the PhD/GRE figure above, since GPA has much lower missingness than the GRE fields in this dataset.

3. Shortcomings of inferring decision_date from status text and term

Year ambiguity from term alone: The term field (e.g. "Fall 2024") gives only a single year, but decisions for a given term can be announced across a real calendar span that crosses a year boundary (e.g. late acceptances or notifications issued in the following January). Using the term year as a flat stand-in for the decision year can misattribute the true calendar year in edge cases.

No date exists for Waitlisted/Interviewed outcomes: The source data only provides acceptance_date and rejection_date columns, so decision_date is structurally unavailable for Waitlisted and Interviewed applicants even though a decision date may exist in reality — it is simply not captured in the schema.

No cross-validation against date_added: The pipeline does not check whether a parsed decision_date falls after date_added. Self-reported entries with data-entry errors could produce an implausible (even negative) days_to_decision without being flagged.

4. Correlation analysis: GPA and GRE-related measures

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GRE vs GRE V (Pearson): r = 0.6727, p = 4.9234e-140
GPA vs GRE (Spearman): r = 0.4292, p = 2.8559e-57
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Both correlations are highly statistically significant and, after excluding out-of-range GRE sentinel values (a placeholder such as 999 used in the source data for "not reported"), both are practically meaningful as well. The GRE/GRE V relationship ($r \approx 0.67$) is a moderately strong positive correlation, which makes intuitive sense since the Verbal subscore is itself a component of the combined GRE total — roughly 45% of the variance in one is shared with the other. The GPA/GRE relationship ($r \approx 0.43$) is a moderate positive monotonic association: applicants with higher undergraduate GPAs tend to also report higher GRE scores, though the relationship is far from perfect, indicating GPA and GRE still capture meaningfully different aspects of an applicant's academic profile rather than being redundant measures of the same thing.

5. Do Accepted and Rejected applicants differ in GPA?

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Test used: Welch's t-test (unequal variances)
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Accepted GPA: mean = 3.7603, n = 7,291
Rejected GPA: mean = 3.7822, n = 8,219
Test statistic: t = -5.4654, p = 4.6946e-08
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The difference is statistically significant, but the direction is counterintuitive: Rejected applicants had a very slightly *higher* mean GPA (3.7822) than Accepted applicants (3.7603). The practical gap is small (about 0.02 GPA points), and with sample sizes in the thousands, even a negligible difference will register as statistically significant. The more reasonable interpretation is that GPA alone is not a strong differentiator between accepted and rejected applicants in this dataset — other unobserved factors (fit, letters of recommendation, statement of purpose, program-specific competitiveness) likely play a larger role than GPA in the outcome.

6. Chi-square test: Degree vs US/International

Degree	American	International
Master's	5,146	2,789
PhD	10,572	10,879

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Chi-square statistic = 563.6234, p = 1.3690e-124, dof = 1
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The association between Degree and US/International status is highly significant. Master's applicants in this cleaned dataset skew notably American (5,146 American vs. 2,789 International, roughly 65/35), while PhD applicants are almost evenly split (10,572 American vs. 10,879 International, roughly 49/51). This indicates Degree type and applicant nationality are not independent — PhD programs in this dataset draw a substantially more international applicant pool than Master's programs do. Note that the "Other" category defined as a valid US/International value never appears in the cleaned data, so the test reduces to a 2x2 table (dof = 1) rather than the 3-category table the schema allows for.

7. Remaining data quality issues after cleaning

Severe GRE missingness, compounded by sentinel/placeholder values: GRE, GRE V, and GRE AW remain roughly 92–94% missing even after cleaning, reflecting true non-reporting by applicants. In addition, the source data used an out-of-range value (999) as a placeholder for "not reported" rather than leaving the field null, which was identified and corrected during cleaning by treating any GRE/GRE V value outside the real scale as missing. This further shrinks the usable sample for GRE-based analysis (e.g. n = 322 for Q1 above) but produces values that are actually trustworthy rather than inflated by placeholder noise.

Unverified self-reported data: The underlying Grad Cafe data is self-reported by applicants with no independent verification, so outcomes, dates, GPA, and GRE scores may contain inaccuracies or exaggerations that no amount of downstream cleaning can catch.

No deduplication: The pipeline does not check for duplicate or repost entries (e.g. the same applicant posting their result more than once), which could inflate counts for specific programs or outcomes.

Incomplete category coverage: The "Other" value for US/International, though defined as valid in the assignment schema, does not appear in the cleaned dataset at all, suggesting either no applicants selected it or upstream data collection never captured it.

Partial application_season coverage: The Early/Mid/Late Cycle buckets only cover September–May; entries with date_added in June–August fall through to "Unknown," which may undercount a real (if smaller) portion of the application cycle.